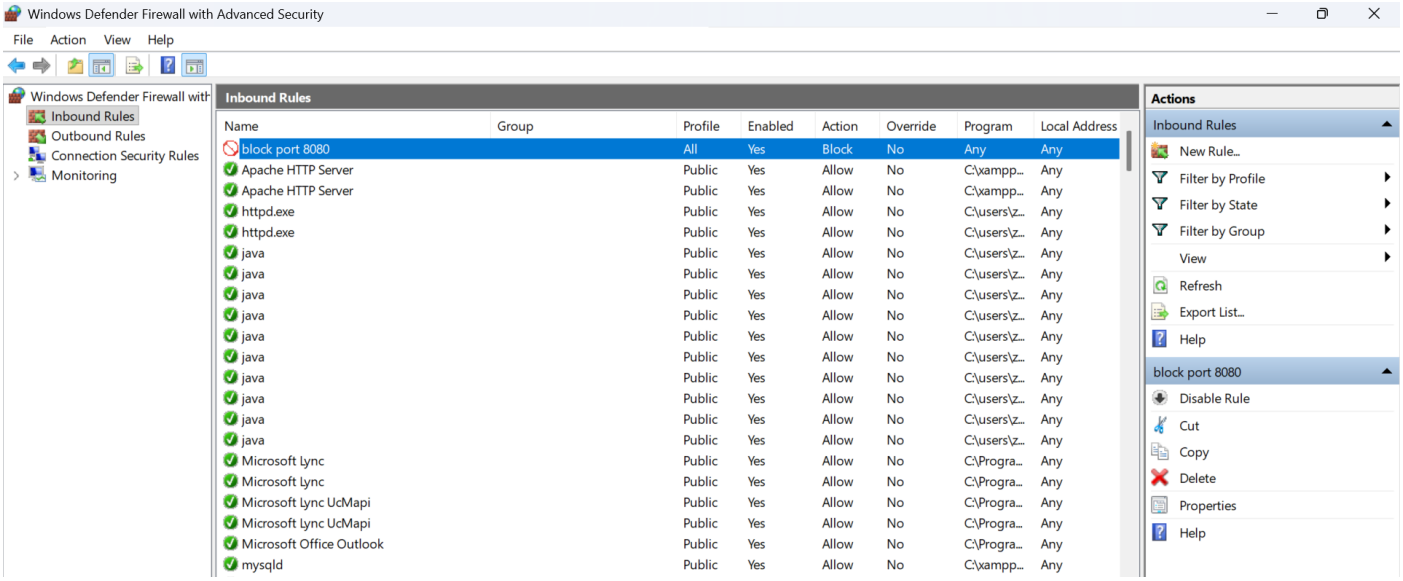
N+ Network Security

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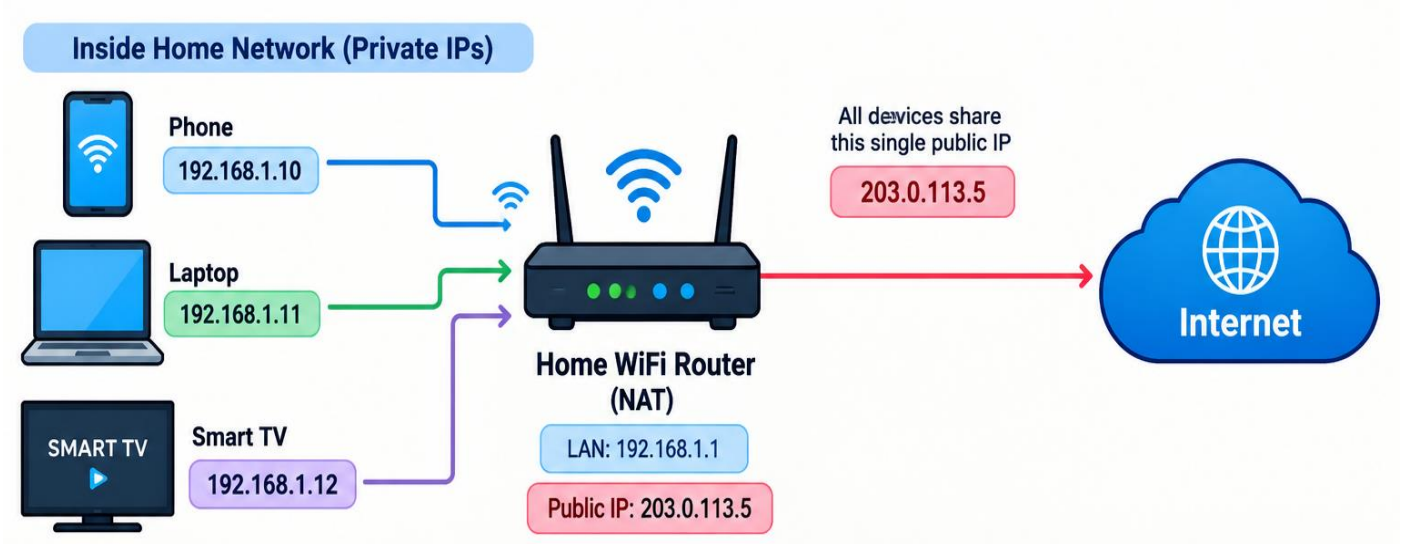
Assignment-2

Prepare By
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1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home Wi-Fi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.Hint:Use any drawing tool or even hand-draw and click a photo.



NAT Translation Table (Example)

Inside Local IP (Private)	Inside Port (Used by Device)	Outside Global IP (Public)	Outside Port (Used on Internet)
192.168.1.10	52345	203.0.113.5	40001
192.168.1.11	52346	203.0.113.5	40002
192.168.1.12	52347	203.0.113.5	40003

How it works (Step by Step)

1. Your device (phone/laptop/TV) sends a request to the internet (e.g., a website).
2. The router replaces the device's private IP with its public IP (**203.0.113.5**) and uses a unique port number to keep track of the connection.
3. The request goes to the internet using the public IP.
4. The response comes back to the router, which looks up the port number and forwards the data to the correct device using its private IP.

This is **NAT (Network Address Translation)** – it lets multiple devices in your home network share one public IP address.

NAT (Network Address Translation):

NAT is a process in which a Wi-Fi router translates private IP addresses into a public IP address. Multiple devices, such as a phone, laptop, and smart TV, can share one public IP address to access the internet. The router keeps track of connections and forwards responses to the correct device.

3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Configure Port Filtering – Block Port 25 (SMTP)

Steps:

1. Open your Wi-Fi router's admin panel (e.g. 192.168.1.1).
2. Login with your username and password.
3. Go to Security / Firewall / Access Control.
4. Find Port Filtering settings.
5. Select Outbound Traffic.
6. Enter port number 25 (SMTP).
7. Select Block / Deny.
8. Click Save / Apply.

Explanation :

Blocking outgoing traffic on port 25 helps prevent spam emails from being sent through our network. It also reduces the risk of infected devices sending unwanted emails to others.

4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.Hint: Focus on source/destination IPs and ports in the rule set.

Firewall Rule: Block port 80 and allow all other ports.

No.	Protocol	Source IP	Destination IP	Port	Action
1	HTTP	192.168.1.5	172.217.166.78	80	BLOCK
2	SSH	10.0.0.10	34.201.23.1	22	ALLOW
3	DNS	192.168.1.7	8.8.8.8	53	ALLOW

Explanation:

1. HTTP uses port 80, which is blocked by the firewall.
2. SSH uses port 22, which is allowed by the firewall.
3. DNS uses port 53, which is allowed by the firewall.

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Answer: 3 Ways Instagram or Zomato Use Firewalls

1. Block Unauthorized Access: Firewalls block unknown users from accessing company servers.

2. Filter Network Traffic: Packet filtering allows trusted traffic and blocks unwanted traffic using IP addresses and port numbers.

3. Prevent Attacks: Firewalls help protect servers from hackers and harmful network traffic.